

RESEARCH MEETING · 2026-09-25

COAST Reproduction Study

From Reproduction Gap to Mechanistic Diagnosis

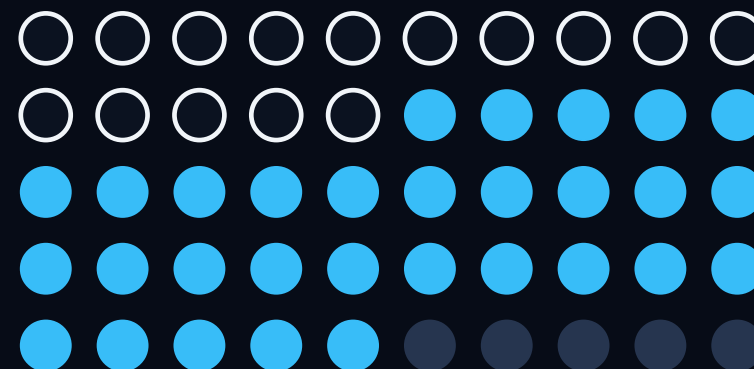
π 0.5 · LIBERO-10 · KS3 and Cross-Task Validation

Stanley Chueh

Reproducing arXiv:2605.17144 v1 — Contrastive Conceptor Activation Steering (COAST)

Paper authors: Miao, Kim, Yang, Ungar

UPDATED THROUGH PHASE 7A



50 LIBERO initial states per task

○ fit / selection · states 0–14

● held-out test · states 15–44

● unused · states 45–49

One question, three layers

“Can we independently reproduce the LIBERO results reported by COAST?”

1 Reproduction

Do the released code and the paper protocol give the reported KS3 numbers?

Gap measured · 5A–5C

2 Mechanism

What does the intervention actually do to the hidden state?

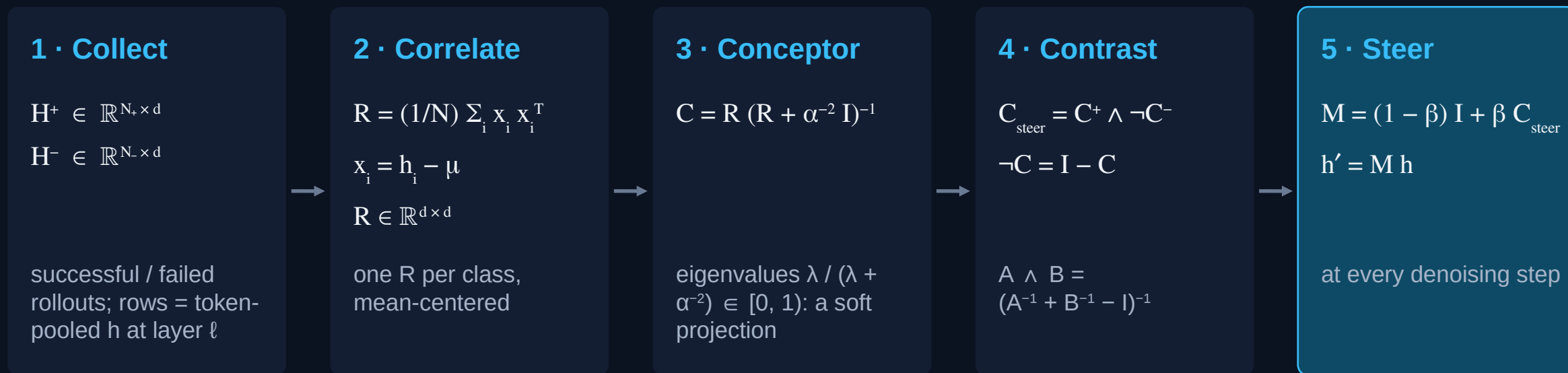
Contraction-dominated · 1E–3A

3 Generalization / Improvement

Does the pattern hold on other tasks — and can the method be improved?

3 more tasks · 7A · method: future

COAST in one slide



$d = 1024$ (action-expert width) · 10 action tokens · 10 Euler denoising steps · layer $\ell \in \{0, 5, 11, 17\}$ of 18

Strategies

Global — one $\mathbf{C}_{\text{steer}}$ for all steps

Per-step — one $\mathbf{C}$ per step

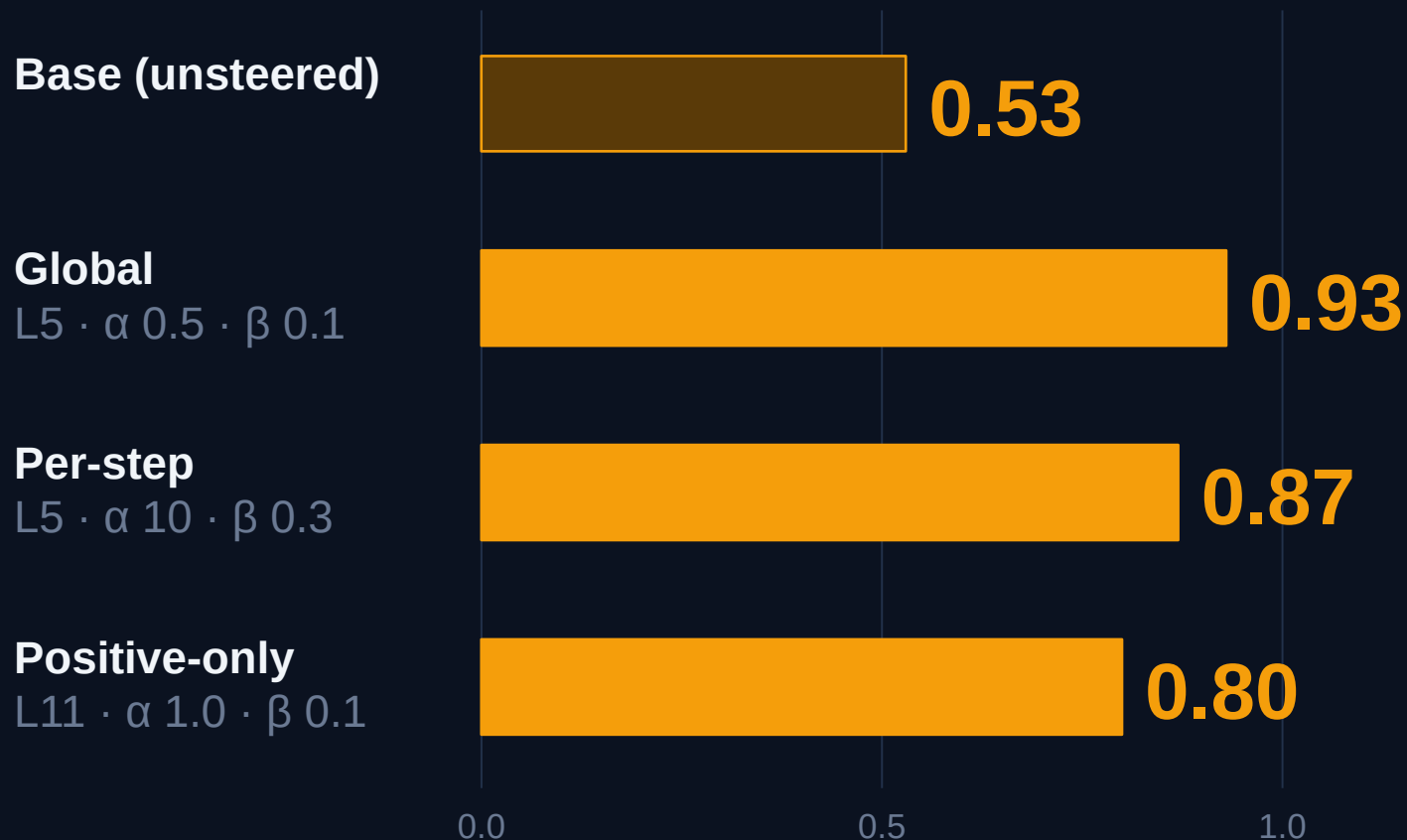
Positive-only — $\mathbf{C} = \mathbf{C}^+$

Paper description. Implementations differ on several of these steps (slides 11, 15).

Source: Paper Sec. 3, App. A.7–A.9 [P]. Released code: `src/openpi/serving/{conceptors,steering}.py`

KS3, Table 4: steering lifts 0.53 → 0.93

PAPER-REPORTED VALUES



How to read these numbers

- Paper text: 30 held-out test rollouts $\Rightarrow$ **0.93 = 28/30**
- Every Table 4 cell is also k/15 $\Rightarrow$ **0.93 = 14/15** (slide 13)
- Base **0.53 = 8/15** equals the Table 21 fit-set rate

LIBERO-10 mean (10 tasks)

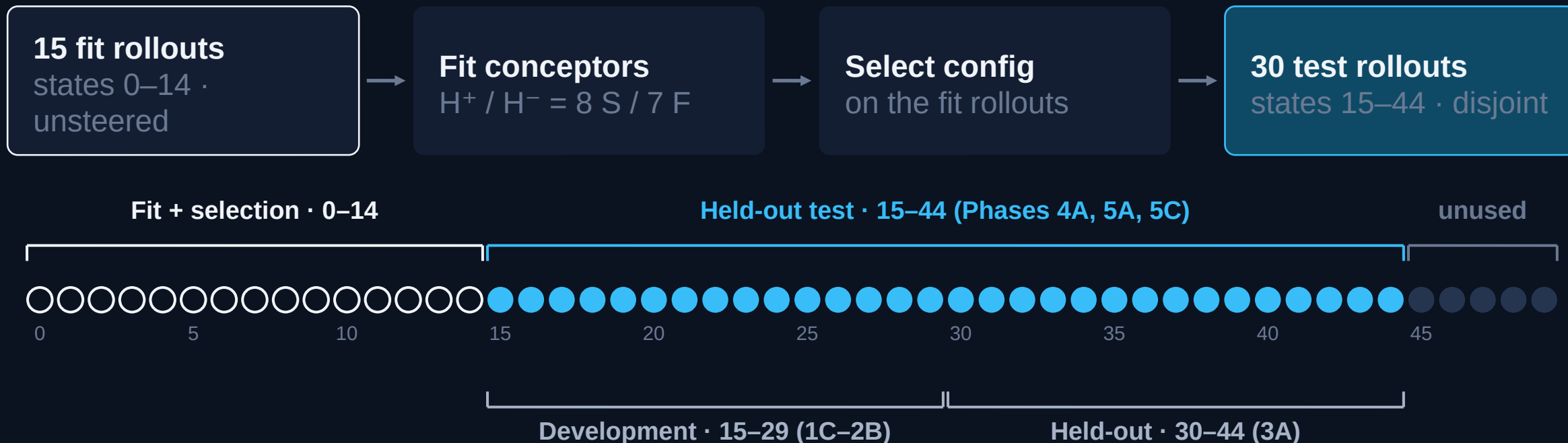
Base 0.43 · Global 0.76

Per-step 0.80 · Positive-only 0.63

KS3 success rate · π 0.5 · LIBERO-10 · Table 4 (Base: Table 1)

Source: Paper Tables 1, 4, 17, 21 [P]; denominators: Phase 6C §5

Reproduction protocol: fit, select, test on disjoint states



LIBERO init state index · KS3 · episode k → state $(\text{seed} + k) \bmod 50$

Our interpretation. The paper lists no init-state IDs or seeds for either split.

Source: Checkpoint openpi-libero-2000 @ aaeabc · KS3 = libero_10 task 2 · COAST main @ 2afa10e · max 520 steps, replan 5 · LIBERO d63b117, robosuite 1.4.0, MuJoCo 3.2.3 · 2× RTX 4090

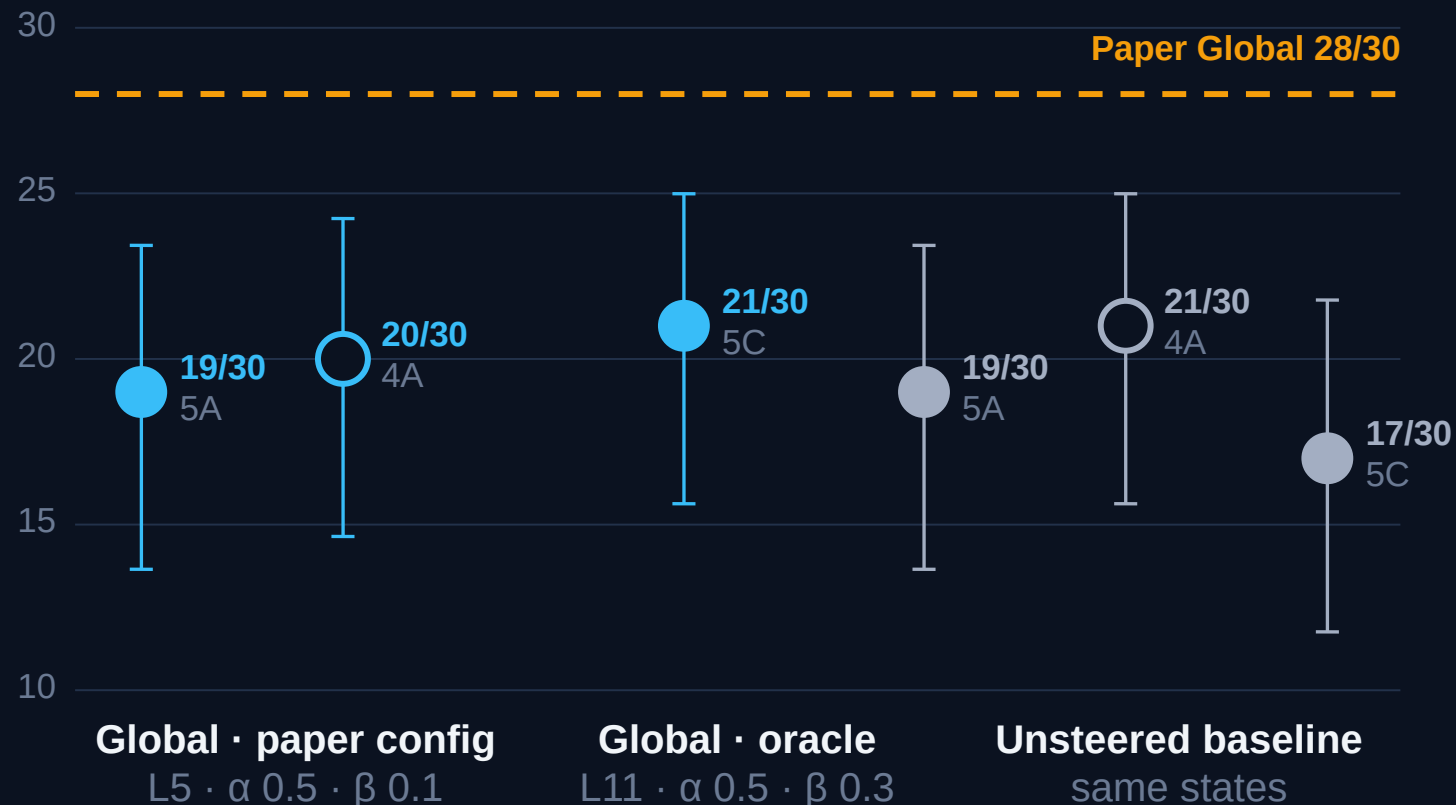
Phase timeline: Phase 0 → 7A



Held-out KS3: 19–21/30 vs the reported 28/30

● released default, unseeded ○ paired noise, seed 100 (Phase 4A)

Successes / 30 · held-out states 15–44 · whiskers: Wilson 95%



Oracle selection did not close the gap.

- Global runs 19, 20, 21 of 30 → pooled **60/90**
- Unsteered runs 19, 21, 17 of 30 → pooled **57/90**
- vs 28/30: Fisher $p = 0.010$ (5A), 0.042 (5C) — reference only, paper states unknown
- Other 5C oracles: positive-only 17/30 (paper 24/30) · per-step 15/30 at $\alpha = 1$ (paper 26/30 at $\alpha = 10$, not like-for-like)

Source: phase5a_ks3_episode_results.csv · phase4a_episode_results.csv (set S) · phase5c_test_episode_results.csv · released main, V0 builder

The unsteered policy alone moves by ± 3 of 15

○ paired noise seed (1D: 100/200/300; 4A: 100)

● released default, unseeded

States 15–29

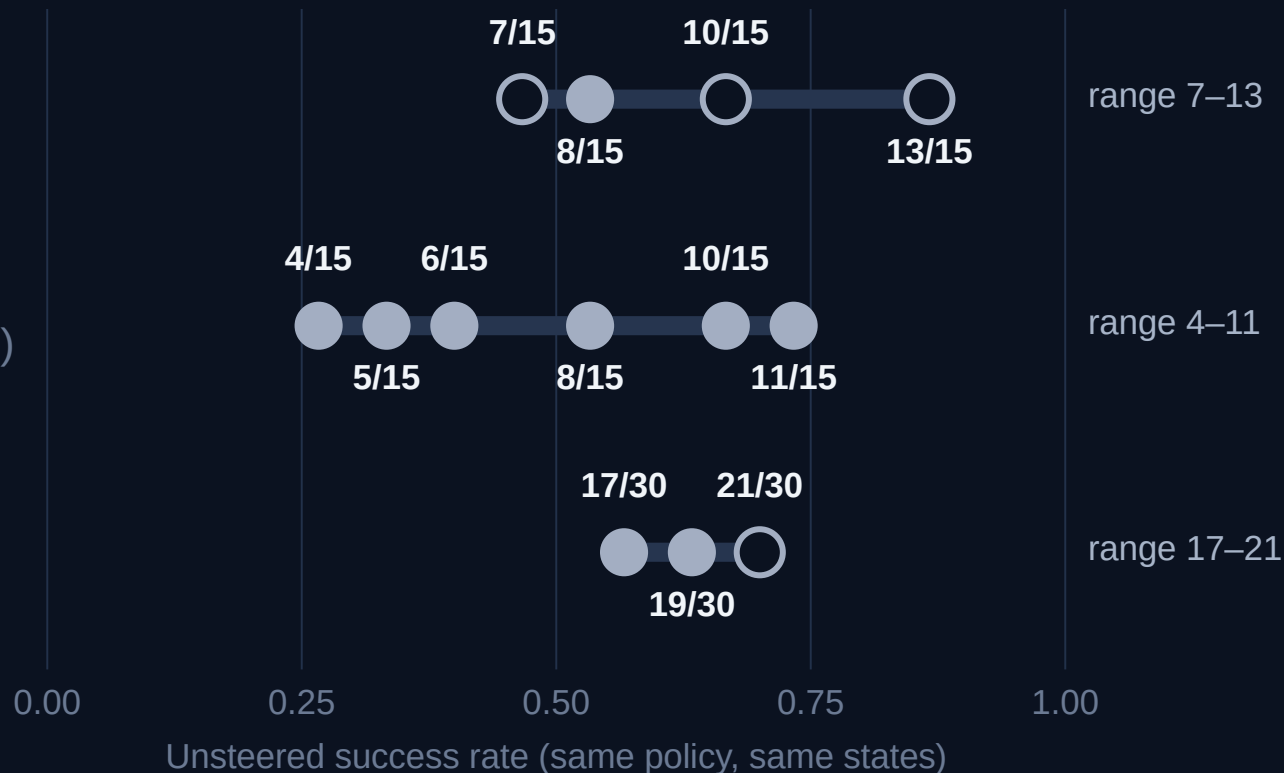
15 episodes

States 0–14

15 episodes (fit states)

States 15–44

30 episodes



A single stochastic run cannot resolve a 2–4 episode difference.

- 11 / 15 states flip outcome with the noise seed alone (1D)
- Binomial SD at $N = 30$, $p \approx 0.6$: ≈ 0.09 (≈ 2.7 episodes)
- Paired noise: $V0 \leftrightarrow V3$ agreement 9/15 $\rightarrow$ 15/15

Source: phase1c/1d_episode_results.csv · phase1a_activation_collection.yaml · phase5c_ks3_oracle_sweep.yaml (fit baselines) · phase4a/5a/5c episode CSVs

What does COAST actually do to h ?

$$M = (1 - \beta) I + \beta C$$

code: $h' = h M^T$ (row) $\Leftrightarrow h' = M h$ (column)

$$h' = (1 - \beta) h + \beta C h$$

$$\Delta h = h' - h = \beta (C h - h)$$

$$\rho = h^T C h / \|h\|^2 \approx 0 \Rightarrow h' \approx (1 - \beta) h$$

$h \in \mathbb{R}^{1024}$ per action token · $M, C \in \mathbb{R}^{1024 \times 1024}$

applied to all 10 tokens at all 10 denoising steps, layer 5

C symmetric $\Rightarrow \| \beta C h \| \leq \beta \lambda_{\max} \|h\|$; $\lambda_{\max} = 0.352 \Rightarrow \leq 3.5\%$ of $\|h\|$

0.0994

$\|\Delta h\| / \|h\|$
 $\beta = 0.1$

-0.9996

$\cos(\Delta h, h)$
 $-1 = \text{pure shrink}$

≈ 0.006

$\rho = h^T C h / \|h\|^2$
trace/d = 0.0185

9,220 hook applications · V0 · L5 α 0.5 β 0.1 · states 15–29 (1E)



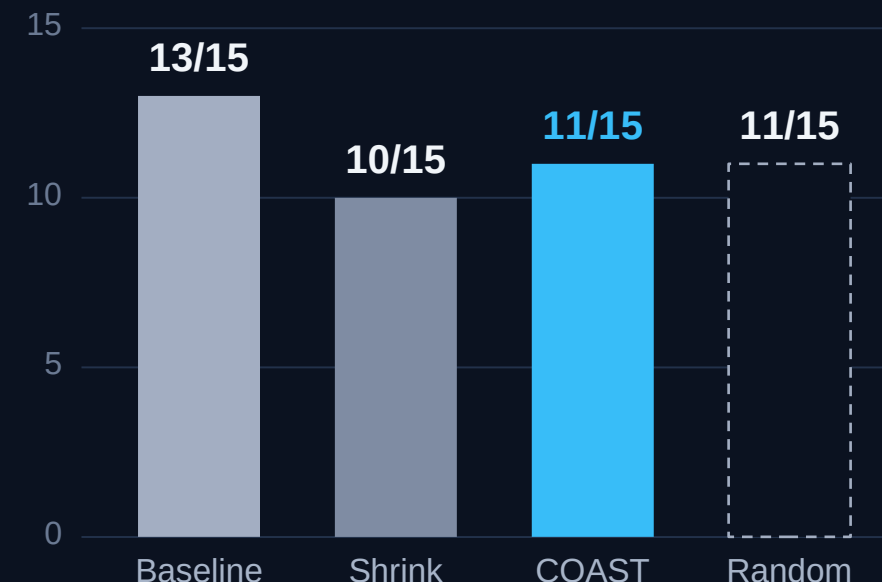
sketch, not to scale: the $\beta C h$ component is magnified

The tested intervention is nearly antiparallel to h — consistent with dominant contraction, $h' \approx 0.9006 \cdot h$.
7A, candidate builder, 4/4 tasks: $\cos(\Delta h, h)$ from -0.9979 to -0.9996

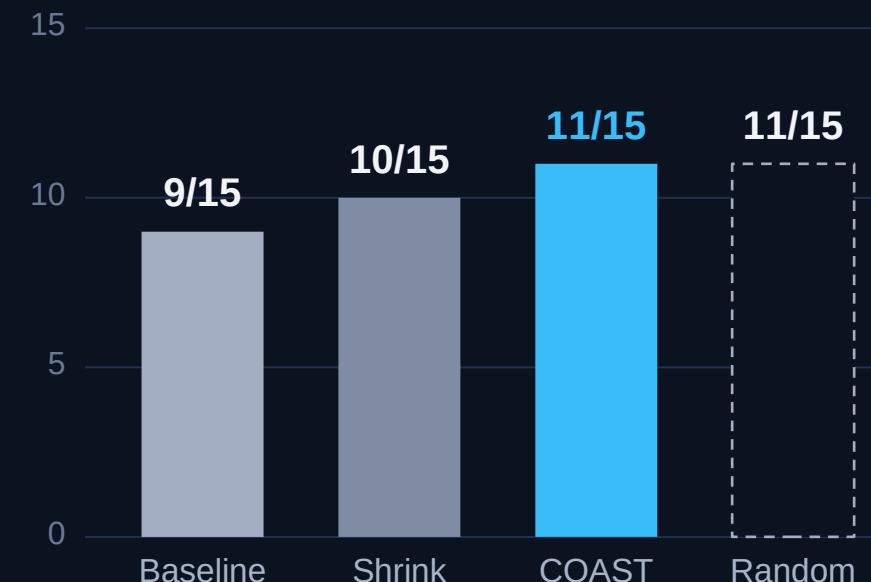
Source: phase1e_intervention_diagnostics.yaml (V0, L5 α 0.5 β 0.1, states 15–29, seed 100) · phase7a_operator_diagnostics.csv · steering.py:447, 461

COAST vs shrinkage vs a spectrum-matched random C

Development · states 15–29 (2A/2B)



Held-out · states 30–44 (3A)

Realized $\|h'\| / \|h\|$ Shrinkage **0.8984**COAST **0.9006**Random **0.9023**

Across all 30 states, real COAST was never the only condition that differed from the other steered conditions.

COAST ↔ shrink **14/15** · COAST ↔ random **15/15**COAST ↔ shrink **14/15** · COAST ↔ random **13/15**

In the tested KS3 setting, the behavioral effect was not clearly distinguishable from shrinkage or a matched random control.

Paired noise (seed 100) · one run per condition · 15 states per split · all exact McNemar $p \geq 0.25$ · Shrink: $M = 0.9 \cdot I$ · Random: C_{real} eigenvalues, random eigenvectors

Source: phase2a/2b_episode_results.csv (states 15–29) · phase3a_episode_results.csv (states 30–44) · $V_0, L_5 \propto 0.5 \beta 0.1$

Paper description $\neq$ released main $\neq$ historical pipeline

Per-step $\alpha = 10$ cannot be expressed in main

Steering merged to main on Jun 1 — after submission

`best_configs.json` is an empty placeholder

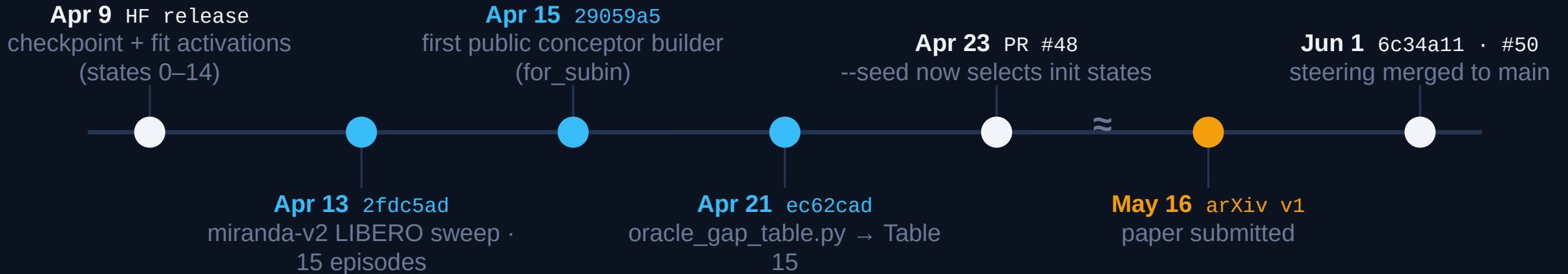
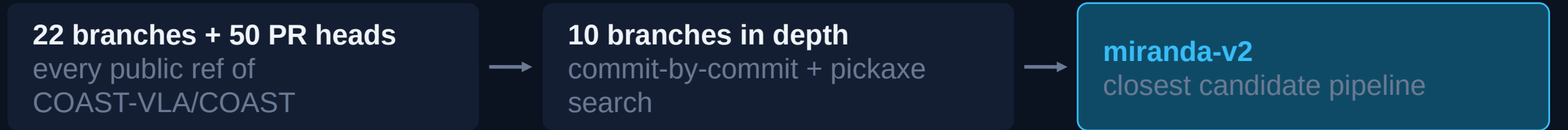
Paper Base = fit-set rate in 10/10 tasks

	Paper text	Released main	miranda-v2
Token pooling	mean over 10 action tokens	✗ every token a row	✓ mean-pooled
Centering	mean-centered R	✗ uncentered	✓ centered
Contrastive formula	$C^+ \wedge \neg C^-$ (pinv AND)	✗ $A(A + B - AB)^{-1}B$	✗ $C_s(I - C_f)$
Per-step α	$\alpha = 10$ (KS3)	✗ fixed $\alpha = 1$	! label a10.0, math $\alpha = 1$
Episodes per evaluation	30 test rollouts	! configurable (docs: 15)	✗ 15 (NUM_EPISODES)
State semantics	different seeds / initial conditions	✓ (seed + k) mod 50	✗ <code>initial_states[k]</code> , $k = 0 \dots 14$
Evaluation split	15 fit + 30 held-out	✓ possible via <code>--seed offset</code>	✗ evaluated on fit states 0–14

✓ as described in the paper · ✗ differs · ! partial / label only

Source: Phase 1A/1B (builder D1–D3), 4B §3, 5B, 6C §4 · miranda-v2 builder = `public for_subin/build_conceptors.py @ 29059a5`

Searching every public ref for the table generator



● miranda-v2 lineage
 ● paper
 ● release / main

The pipeline that plausibly generated Table 4 predates both the held-out seeding change (#48) and the released code.

Source: phase6c_branch_matrix.csv · phase6c_author_branch_archaeology.md §2–3 · public git history of COAST-VLA/COAST

Why miranda-v2 looks like the table generator

STRONG PROVENANCE EVIDENCE — NOT DEFINITIVE PROOF

152

configurations per task

Paper: “up to 152 configurations” = the miranda-v2 sweep: 135 steered + 9 random + 8 positive-only

18

“Selected Grid” ($\pi 0.5$ LIBERO)

= the count `oracle_gap_table.py` derives from the sweep's condition names

a10.0

per-step “ $\alpha = 10$ ”

A loop label in the condition name `per_step*_L5_a10.0_b0.3`; α does not enter the per-step concepthor

10 / 10

positive-only cells

All inside the sweep grid $L\{5,11\} \times \alpha\{0.5,1\} \times \beta\{0.1,0.3\}$ (8 conditions), not Table 14's 60

k / 15

30 / 30 Table 4 cells

If $N = 30$: $P(\text{all 30 numerators even}) = 4.0 \times 10^{-9}$. Control: Table 8 (16-env pipeline) shows $k/16$

Caption

Table 15

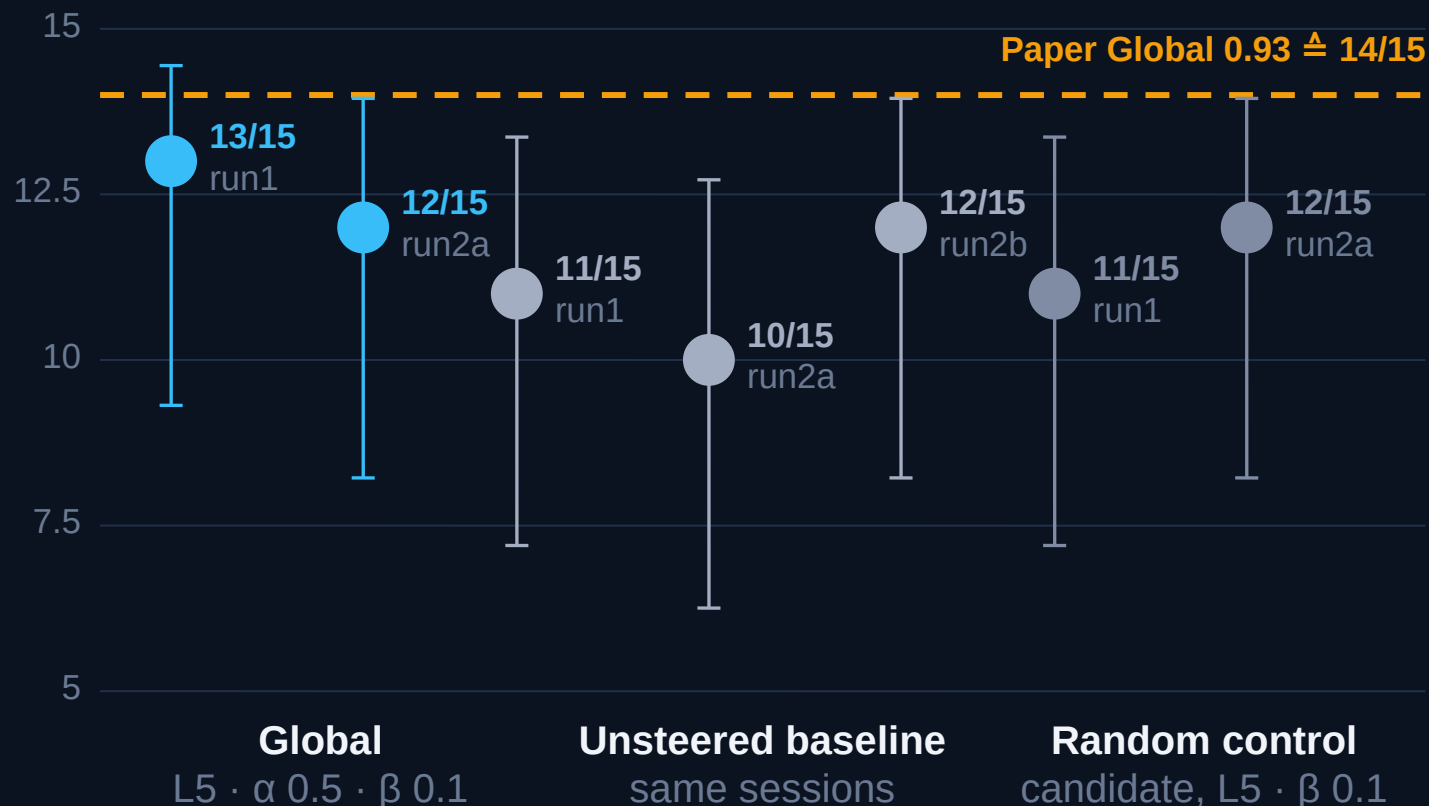
Near-verbatim rendering of the LaTeX f-string in `oracle_gap_table.py`, same benchmark order

Source: Paper Tables 4, 8, 14, 15 [P] · miranda-v2 2fdc5ad / ec62cad [R] · phase6c_author_branch_archaeology.md §1, §5

miranda-v2, run as written: 12–13/15 on the fit states

HISTORICAL FIT-STATE SEMANTICS · NOT HELD-OUT

Successes / 15 · states 0–14 (= fit states) · whiskers: Wilson 95%



The historical pipeline approximately reproduces the numerical Global score, but baseline and random controls reach the same range.

- Pooled: Global **25/30** · baseline **33/45** · random **23/30**
- Fisher: vs baseline $p = 0.40$ · vs random $p = 0.75$ · vs 14/15 $p = 0.65$
- Per-step cell as coded: 9/15, 9/15 (paper 13/15) · Positive-only 7/15 (paper 12/15)
- Evaluated states 0–14 are the authors' fit-rollout states

Source: phase6c_candidate_test_results.csv · miranda-v2 @ 29059a5 (unmodified) · authors' fit activations · states 0–14, unseeded

Paper vs public code, component by component

Component	Paper (text)	Released main · 2afa10e	miranda-v2 · 29059a5
Evaluation episodes	30 per condition	! configurable (docs: 15)	✗ 15 (NUM_EPISODES = 15)
Fit / test separation	15 fit + 30 disjoint test	✓ via --seed offset (post-#48)	✗ none: eval states 0–14 = fit
Token pooling	mean over 10 action tokens	✗ every token is a row	✓ mean-pooled
Centering	mean-centered R	✗ uncentered $R = X^T X / N$	✓ centered
Contrastive operation	$(A^{-1} + B^{-1} - I)^{-1}$, pinv	✗ $A (A + B - AB)^{-1} B$	✗ $C_s (I - C_f)$
Global aggregation	all denoising steps	✓ all steps × all tokens	✗ denoising step 0 only
Per-step α	$\alpha = 10$ (KS3)	✗ fixed $\alpha = 1.0$	✗ label a10.0; conceptor $\alpha = 1$
Per-step mechanism	separate C per step	✓ time-varying	✗ fixed ds-0 or ds-9 C
Grid size	240 (Table 14); text: up to 152	! configurable; per-step α fixed	✓ 152 (matches text + cells)

✓ as in the paper text ✗ differs ! partial / unclear

! miranda-v2 builder column = public for_subin builder (29059a5); the 2026-04-13 builder that wrote the NPZ actually used is not public.

Source: Phase 1A/1B (D1–D3), 4B §3, 5B §2, 6C §4 · paper Sec. 3.2, App. A.8–A.9, Table 14

Does the KS3 pattern generalize?

COMPLETE · 810 episodes

Task selection · paper values only

Eligible: Table 21 $S \geq 3$ and $F \geq 3$; KS3 excluded

A **LR2a** · largest paper gain
 Paper Base 0.40 → Global 0.80
 L11 · α 0.5 · β 0.3

B **KS4** · median paper gain
 Paper Base 0.40 → Global 0.73
 L5 · α 1.0 · β 0.1

C **LR1** · smallest paper gain
 Paper Base 0.80 → Global 0.93
 L11 · α 0.5 · β 0.3

Two arms, one conceptor

Arm H · historical
 states 0–14 · 15 episodes
 = the fit states

Arm T · held-out
 states 15–44 · 30 episodes
 disjoint from the fit states

Conditions

- ☐ Baseline (unsteered)
- ☒ Paper Global (exact config)
- ☐ Random control (candidate)*

2 repeats · 810 episodes · unseeded

* not spectrum-matched

Primary quantity

$$G_H = SR_{\text{COAST},H} - SR_{\text{BASE},H}$$

$$G_T = SR_{\text{COAST},T} - SR_{\text{BASE},T}$$

$$PGG = G_H - G_T$$

$PGG > 0$: the apparent gain is larger under historical fit-state evaluation than under held-out evaluation.

Case rule (pre-registered)

A: $G_H > G_T$ in most tasks, no clear held-out advantage

B: mixed · C: COAST beats baseline and random on T

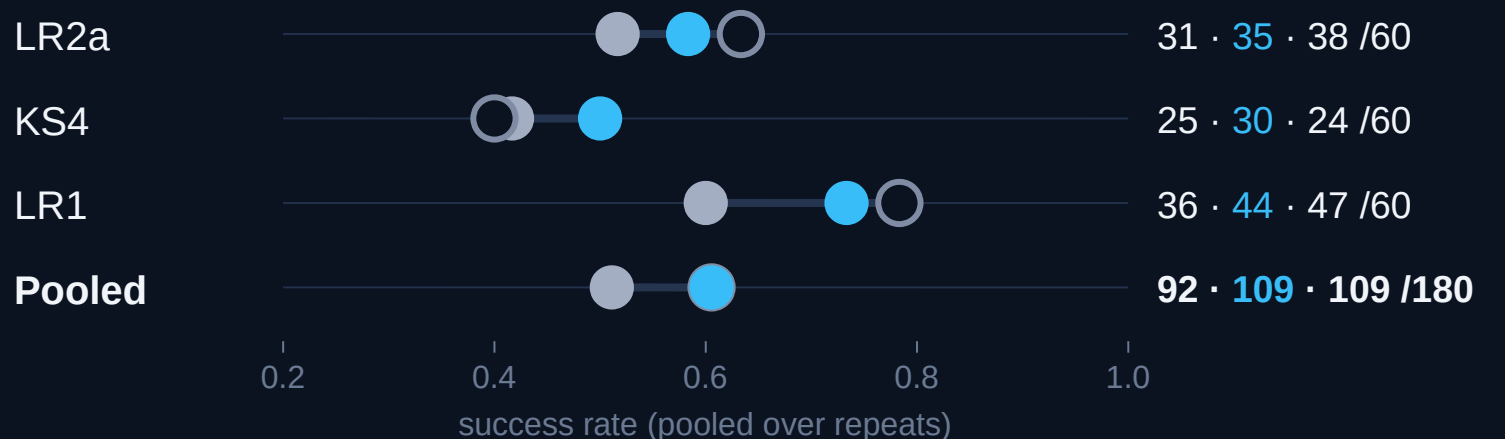
clear = pooled Fisher $p < 0.05$ and same sign in both repeats

Held-out: paper Global $\approx$ random control in every task

Arm H · states 0–14 (fit states) · 2×15

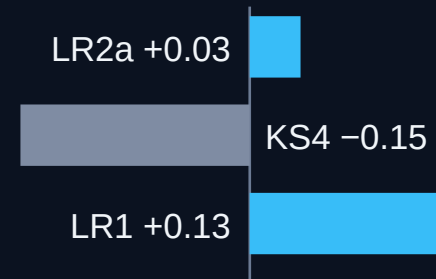


Arm T · held-out states 15–44 · 2×30



● Baseline ● Paper Global ○ Random control | Paper Global (as /15)

$$PGG = G_H - G_T$$



- Pooled held-out: Global **109/180** = Random **109/180**; baseline 92/180 (CMH $p = 0.086$)
- No task met “clearly better” (pooled $p < 0.05$ and same sign in both repeats)
- PGG not consistent: mean 0.006 $\rightarrow$ no systematic fit-to-held-out drop
- Pre-registered classification: **Case A**

Repeats share init states $\rightarrow$ pooled counts are not independent · candidate random concepor is not spectrum-matched (quota ≈ 0.18)

Source: phase7a_summary.csv · phase7a_analysis.json · phase7a_cross_task_validation.yaml

What we can and cannot conclude

SUPPORTED

1. Released main does not reproduce the reported KS3 Global value under our held-out protocol (19–21/30 vs 28/30).
2. miranda-v2 approximately reproduces the KS3 Global number under its native 15-fit-state semantics (12–13 vs 14/15) — with controls in the same range; not reached for LR2a / KS4 (7A).
3. No public implementation matches the full paper-described protocol.
4. In 4/4 tasks the paper Global configuration shows no held-out advantage over a random conceptror.

OPEN

- Original conceptror NPZ (HF repo returns HTTP 401)
- Exact 2026-04-13 conceptror builder
- Original per-condition summary.json files
- 45-episode rerun outputs
- Checkpoint identity (training run vs released HF weights)
- Policy-noise seeding and runs per table cell

NOT CLAIMED

- That the paper is invalid
- That results were fabricated
- That COAST universally fails

What we asked the author

STATUS · REPLY PENDING

1 Table source

Were Tables 4/17 generated from the 15-episode miranda-v2 sweep (best condition per strategy from `summary.json`)?

2 Held-out test

Was a separate 30-rollout held-out evaluation run — e.g. the `steering_results_45ep` reruns — and on which states?

3 Per-step α

Is the KS3 per-step “ $\alpha = 10$ ” a true $\alpha = 10$ conceptr, or the loop label of `per_step_{0|9}_L5_a10.0_b0.3`?

4 Builder & checkpoint

Which builder / NPZ produced the reported results, and is `libero_b200_bs512/2000` = HF `openpi-libero-2000`?

We ask which protocol the tables report — not whether the results are correct.

Source: `phase6b_author_questions.md` · `phase6c_author_branch_archaeology.md` §10

Next: Phase 7B, mechanism first

PLANNED · PHASE 7B

- Phase 7B (mechanism-first): is a directional, norm-preserving component what the operator lacks?
- Same disjoint held-out protocol (states 15–44), repeats, random and shrinkage controls
- Pre-registered power target: 60 episodes/arm resolves only $|\Delta| \gtrsim 0.2\text{--}0.25$

Success criterion for any future method: beat baseline **and** matched random / shrinkage controls on disjoint held-out states, with repeats.

FUTURE WORK · NOT YET TESTED

If the mechanism holds...

Current $h' = (1 - \beta) h + \beta C h \approx (1 - \beta) h$

Possible $h' = h + \gamma D(h)$

- preserve the hidden-state norm ($\|h'\| \approx \|h\|$)
- isolate a success-specific direction $D(h)$
- avoid uniform contraction
- evaluate on held-out states against matched controls

A direction for discussion — no such method has been implemented or tested.

- 1 Under a documented held-out protocol, the released implementation shows a reproducible gap to the reported KS3 result: 19–21/30 vs 28/30.
- 2 Historical-code archaeology explains much of the numerical discrepancy — the closest author pipeline scores 15 fit-state rollouts — and exposes protocol and implementation differences from the paper text.
- 3 Cross-task validation found no held-out advantage over a random concepthor in 4/4 tasks; next is a mechanism-first Phase 7B before proposing any new steering method.